Vendor: McMaster-Carr
Edge Diameter: ⅜” diameter of threads
Minor Diameter of ⅜”-16 Threads: 0.3006
Thread length ¼”

Part Type	Part #	Diameter (in)	Thread	Length (in)	Yield Strength (kpsi)	Cost	Material	Factor of Safety
Threaded Rotary Shaft	4166N17	½”	⅜”-16	12”	58	$28.65	1117 Carbon Steel	0.0775
Threaded Rotary Shaft	4166N24	¾”	½”-13	12”	58	$37.33	1117 Carbon Steel	0.0775
Threaded Rotary Shaft	4166N31	1”	⅝”-11	12”	58	$53.95	1117 Carbon Steel	0.0775
Rotary Shaft	1346K31	¾”	N/A	12”	87	$15.39	1566 Carbon Steel	
Rotary Shaft	1886K41	¾”	N/A	12”	65.3	$11.26	1045 Carbon Steel	

12L14 Carbon Steel	60.2 kpsi
4140 Alloy Steel	60.2 kpsi

```
clear
format compact
%McMaster-Carr Products
RPMs = 30000;%Angular Velocity
P = 11.4; %hp
Sys = 1000; %kpsi ...material???
D = 0.3006; %Chosen Diameter
J = pi*D^4/32; %Moment of Inertia
T = (5252*P)/RPMs; %Torque in*lbf
tmax = 0.5*D*T/J; %Shear Stress psi
nd = 0.5*Sys/tmax %factor of safety
```

Sources

- https://www.azom.com/article.aspx?ArticleID=6562
- https://blog.engineeringpaper.xyz/calculating-drive-shaft-torque
- https://www.machiningdoctor.com/threadinfo/?tid=38